Assignment 2

August 15, 2026

1 Part 1

For a particle of mass m moving with a velocity v, the action S along a path is given by the integral of the Lagrangian. Since the potential energy here is 0, the kinetic energy is given by:

$$\frac{1}{2}mv^2$$

The Lagrangian is:

$$L = T - V = \frac{1}{2}mv^2$$

The action is:

$$S = \int L dt = \int \frac{1}{2}mv^2 dt = \frac{1}{2}mv^2\left(\frac{s}{v}\right) = \frac{1}{2}mvs$$

Path B is slightly longer than Path A because the neutron travels an extra distance d. in Figure 1, that extra distance forms a right triangle with the nucleus's diameter D and the scattering angle θ, giving

$$d = D\sin\theta$$

Because action depends directly on distance, the difference in action between the two paths is:

$$\Delta S = mvD\sin\theta$$

2 Part 2

Total wave probability amplitude $W_{A\to B}$ is the sum of the wave functions for path A and path B

$$W_{A\to B} = W_A + W_B$$

The probability amplitude for a particle traveling along a specific classical path is:

$$W_A = Ne^{iS_A/\hbar}, \quad w_B = Ne^{iS_B/\hbar}$$

where N is a normalization constant.

The probability P is given by the squared absolute value of the total amplitude:

$$P_{A\to B} = \left(Ne^{iS_A/\hbar} + Ne^{iS_B/\hbar}\right)^* \left(Ne^{iS_A/\hbar} + Ne^{iS_B/\hbar}\right)$$

$$P_{A\to B} = N^2 \left(e^{-iS_A/\hbar} + e^{-iS_B/\hbar}\right) \left(e^{iS_A/\hbar} + e^{iS_B/\hbar}\right)$$

$$PA \to B = N^2 \left(1 + e^{i(S_B - S_A)/\hbar} + e^{-i(S_B - S_A)/\hbar} + 1\right)$$

but $\Delta S = S_B - S_A$:

$$PA \to B = N^2 \left(2 + e^{i\Delta S/\hbar} + e^{-i\Delta S/\hbar}\right)$$

Applying the identity $\cos z = \frac{e^{iz}+e^{-iz}}{2}$:

$$P = 2N^2 \left(1 + \cos \frac{\Delta S}{\hbar}\right)$$

Thus the probability to detect a neutron is proportional to $(1 + \cos \frac{\Delta S}{\hbar})$

3 Part 3

The detection probability depends on the term $(1 + \cos \frac{\Delta S}{\hbar})$. A destructive interference minimum occurs when the cosine term reaches its minimum value of -1. This requires the argument of cosine:

$$\frac{\Delta S}{\hbar} = \pi \Rightarrow |\Delta S| = \pi\hbar$$

At this angle ($\theta \approx 18°$), $\Delta S = \pi\hbar$, meaning the phase difference between the two path amplitudes is π radians (180°). Also, the two phasors associated with paths A and B point in opposite directions, leading to destructive interference.

4 Part 4

At the first minimum angle θ_0, the action difference satisfies $\Delta S = \pi\hbar$[cite: 1]. Substituting for ΔS:

$$mvD \sin\theta_0 = \pi\hbar$$

Solving for D:

$$D = \frac{\pi\hbar}{mv\sin\theta_0}$$

Since $\hbar = \frac{h}{2\pi}$, substituting this gives:

$$D = \frac{h}{2mv\sin\theta_0}$$

Given:

Planck constant $h \approx 6.63 \times 10^{-34} J \cdot s$
neutron mass $m = 1.67 \times 10^{-27} kg$
neutron velocity $v = 10^8 m/s$
angle $\theta_0 = 18°$, $\sin(18°) \approx 0.309017$:

$$D = \frac{6.63 \times 10^{-34}}{2 \times (1.67 \times 10^{-27}) \times (10^8) \times 0.309017} = \frac{6.63 \times 10^{-34}}{1.03218 \times 10^{-19}}$$

$$D \approx 6.42 \times 10^{-15} m$$